

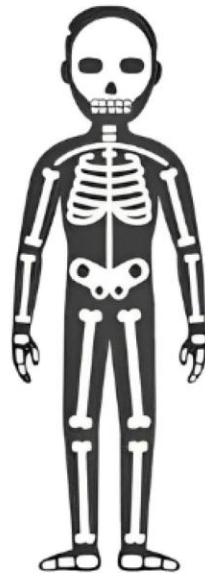
HTML



HTML



HTML the Skeleton



CSS



CSS the Skin



JS



Javascript the Brain



WHAT WE WILL DISCUSS ?

- What is HTML ?
- HTML Meta Tag
- HTML Elements
- Lists in HTML
- Links in HTML
- Images in HTML
- Tables in HTML
- Forms in HTML
- Audio and Video in HTML
- HTML Block and Inline Elements
- Semantic Elements in HTML
- Project Overview

WHAT IS HTML ?

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the **structure of a Web page**
- HTML consists of a series of elements
- HTML elements **tell the browser how to display the content**
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

HTML *<meta>* Tag

```
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Free Web tutorials">
  <meta name="keywords" content="HTML, CSS, JavaScript">
  <meta name="author" content="John Doe">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
```

- Metadata is used by browsers (how to display content or reload page), search engines (keywords), and other web services.
- The `<meta>` tag defines metadata about an HTML document. Metadata is data (information) about data.

Basic HTML Structure

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph



HTML ELEMENTS

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	<i>none</i>	<i>none</i>

HTML Headings

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

HTML ELEMENTS

HTML Paragraphs

Tag	Description
<code><p></code>	Defines a paragraph
<code><hr></code>	Defines a thematic change in the content
<code>
</code>	Inserts a single line break
<code><pre></code>	Defines pre-formatted text

Pre

```
<pre>
My Bonnie lies over the ocean.
My Bonnie lies over the sea.
My Bonnie lies over the ocean.
Oh, bring back my Bonnie to me.
</pre>
```

My Bonnie lies over the ocean.
My Bonnie lies over the sea.
My Bonnie lies over the ocean.
Oh, bring back my Bonnie to me.

hr

```
<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
```

This is heading 1

This is some text.

LISTS IN HTML

ORDERED LIST

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

```
1. Coffee
2. Tea
3. Milk
```

UNORDERED LIST

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

```
• Coffee
• Tea
• Milk
```

DESCRIPTION LIST

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

```
Coffee
  - black hot drink
Milk
  - white cold drink
```

- <dt> tag defines the term (name)
- <dd> tag describes each term

LINKS IN HTML

HTML Links - Syntax

`<a href="url">link text</a>`

Ex : `<a href="https://www.instagram.com/">Instagram</a>`

HTML Links - The target Attribute

The target attribute can have one of the following values:

- `_self` - Default. Opens the document in the same window/tab as it was clicked
- `_blank` - Opens the document in a new window or tab
- `_parent` - Opens the document in the parent frame
- `_top` - Opens the document in the full body of the window

Ex : `<a href="https://www.instagram.com/" target="_blank">Instagram</a>`

LINKS IN HTML

Absolute URLs vs. Relative URLs

<h2>Absolute URLs</h2>

<p>W3C</p>

<p>Google</p>

<h2>Relative URLs</h2>

<p>HTML Images</p>

<p>CSS Tutorial</p>

USE an Image as a Link

Link to an Email Address

Send
email

IMAGES IN HTML

HTML Image - Syntax

```

```

Ex : ``

Image Size - Width and Height

```

```

TABLES IN HTML

A Simple HTML Table

```
<table>
  <tr>
    <th>Company</th>
    <th>Contact</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Alfreds Futterkiste</td>
    <td>Maria Anders</td>
    <td>Germany</td>
  </tr>
  <tr>
    <td>Centro comercial
Moctezuma</td>
    <td>Francisco Chang</td>
    <td>Mexico</td>
  </tr>
</table>
```

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

TABLES IN HTML

Table Cells

td stands for table data

Table Rows

tr stands for table row.

Table Headers

th stands for table header.

<caption> -> Defines a table caption

```
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
    <th>Person 3</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

FORMS IN HTML

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```

First name:

John

Last name:

Doe

<p>Choose your favorite Web language:</p>

```
<form>
  <input type="radio" id="html" name="fav_language" value="HTML">
  <label for="html">HTML</label><br>
  <input type="radio" id="css" name="fav_language" value="CSS">
  <label for="css">CSS</label><br>
  <input type="radio" id="javascript" name="fav_language" value="JavaScript">
  <label for="javascript">JavaScript</label>
</form>
```

Choose your favorite Web language:

- ☐ HTML
- ☐ CSS
- ☐ JavaScript

FORMS IN HTML

```
<form>
  <input type="checkbox" id="vehicle1" name="vehicle1" value="Bike">
  <label for="vehicle1"> I have a bike</label><br>
  <input type="checkbox" id="vehicle2" name="vehicle2" value="Car">
  <label for="vehicle2"> I have a car</label><br>
  <input type="checkbox" id="vehicle3" name="vehicle3" value="Boat">
  <label for="vehicle3"> I have a boat</label>
</form>
```

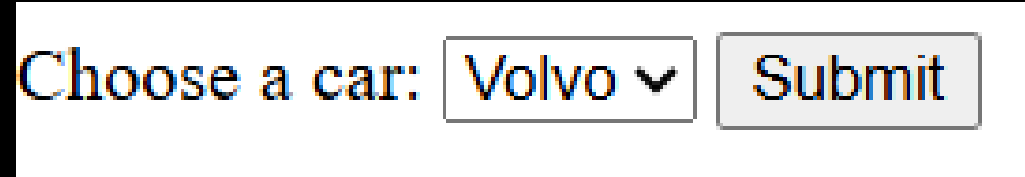
☐ I have a bike
☐ I have a car
☐ I have a boat

```
<form >
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" value="John"><br><br>
  <input type="submit" value="Submit">
</form>
```

First name:

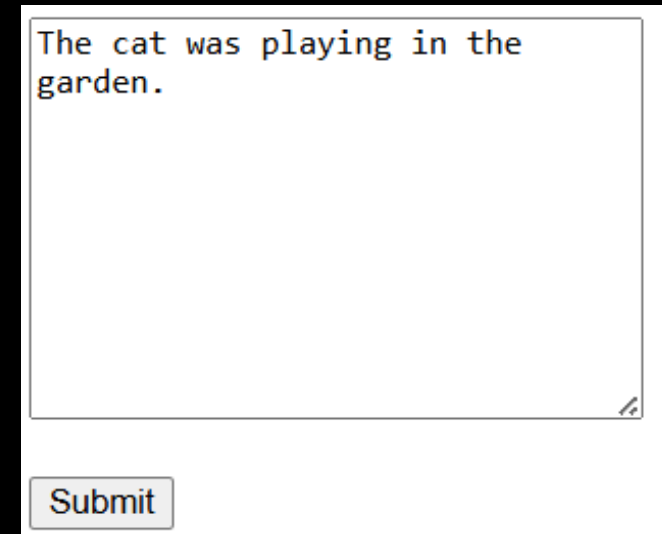
FORMS IN HTML

```
<label for="cars">Choose a car:</label>
<select id="cars" name="cars">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```



Choose a car:

```
<form>
  <textarea name="message" rows="10"
  cols="30">The cat was playing in the
  garden.</textarea>
  <br><br>
  <input type="submit">
</form>
```



The cat was playing in the garden.

FORMS IN HTML

HTML Input Types

- `<input type="button">`
- `<input type="checkbox">`
- `<input type="color">`
- `<input type="date">`
- `<input type="datetime-local">`
- `<input type="email">`
- `<input type="file">`
- `<input type="hidden">`
- `<input type="image">`
- `<input type="month">`
- `<input type="number">`
- `<input type="password">`
- `<input type="radio">`
- `<input type="range">`
- `<input type="reset">`
- `<input type="search">`
- `<input type="submit">`
- `<input type="tel">`
- `<input type="text">`
- `<input type="time">`
- `<input type="url">`
- `<input type="week">`

FORMS IN HTML

HTML Input Attributes

Readonly Attribute

```
<input type="text" id="fname" name="fname" value="John" readonly><br>
```

Placeholder Attribute

```
<input type="tel" id="phone" name="phone" placeholder="123-45-678">
```

Required Attribute

```
<input type="text" id="username" name="username" required>
```

Step Attribute

```
<input type="number" id="points" name="points" step="3">
```

AUDIO AND VIDEO IN HTML

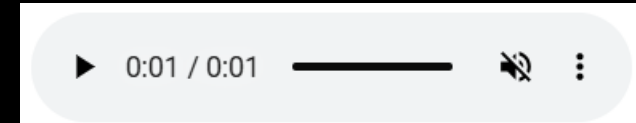
VIDEO IN HTML

```
<video width="320" height="240" autoplay muted>  
  <source src="movie.mp4" type="video/mp4">  
Your browser does not support the video tag.  
</video>
```



AUDIO IN HTML

```
<audio controls autoplay muted>  
  <source src="horse.mp3" type="audio/mpeg">  
Your browser does not support the audio element.  
</audio>
```



YOUTUBE IN HTML

```
<iframe width="420" height="315"  
src="https://www.youtube.com/embed/tgbNymZ7vqY?playlist=tgbNymZ7vqY&loop=1">  
</iframe>
```

HTML BLOCK AND INLINE ELEMENTS

BLOCK ELEMENTS IN HTML

<address>	<article>	<aside>	<blockquote>	<canvas>	<dd>	<div>	<dl>
<dt>	<fieldset>	<figcaption>	<figure>	<footer>	<form>	<h1>-<h6>	<header>
<hr>		<main>	<nav>	<noscript>		<p>	<pre>
<section>	<table>	<tfoot>		<video>			

INLINE ELEMENTS IN HTML

<a>	<abbr>	<acronym>		<bdo>	<big>	 	<button>
<cite>	<code>	<dfn>		<i>		<input>	<kbd>
<label>	<map>	<object>	<output>	<q>	<samp>	<script>	<select>
<small>			<sub>	<sup>	<textarea>	<time>	<tt>
<var>							

SEMANTIC ELEMENTS IN HTML

Tag	Description
<u><article></u>	Defines independent, self-contained content
<u><aside></u>	Defines content aside from the page content
<u><details></u>	Defines additional details that the user can view or hide
<u><figcaption></u>	Defines a caption for a <figure> element
<u><figure></u>	Specifies self-contained content, like illustrations, diagrams, photos, code listings, etc.
<u><footer></u>	Defines a footer for a document or section
<u><header></u>	Specifies a header for a document or section
<u><main></u>	Specifies the main content of a document
<u><mark></u>	Defines marked/highlighted text
<u><nav></u>	Defines navigation links
<u><section></u>	Defines a section in a document
<u><summary></u>	Defines a visible heading for a <details> element
<u><time></u>	Defines a date/time

SEMANTIC ELEMENTS IN HTML

Difference between Semantic and Div

<i>Semantic Tags</i>	<i><div></i>
Have meaning	No meaning
Describe content purpose	Generic container
Better for SEO	Not useful for SEO
Better accessibility	Less accessibility support
Easier to understand structure	Structure not clear
Used for specific sections	Used mainly for styling/layout

SEO (Search Engine Optimization) is the process of improving a website so that it appears higher in search engine results like Google.

HTML5 FEATURES

Semantic Tags

- Introduced tags like <header>, <footer>, <main>, <section>, <nav>
- Improves SEO and accessibility

Audio and Video Support

- Added <audio> and <video> tags
- No external plugins required

New Form Input Types

- Added inputs like email, date, number, range
- Better validation and user experience

Canvas Element

- Added <canvas> for drawing graphics and animations
- Used in games and charts

PROJECT FOR SESSION

Build a personal profile page in pure HTML/CSS:

- A header with your name + photo placeholder
- A short bio section
- A links section

Push it to a public GitHub repo!

We will Add CSS in Next Session.....

Jaswanth Kumar



I am Currently Pursuing DSAI in IIT Bhilai.I am interested in learning new tech stack.

- [kjk_.06](#)
- [Jaswanth-Kumar-2007](#)

THANK YOU

